Rev 2
6/9/26

Design Rulebook

Project Goals

Generate 3D printable electromagnetic circuits in OpenSCAD. Circuits will consist of a protopasta conductive PLA trace layer and an insulating layer that separates the traces.

Physical Laws

- Conductive trace layer resistivity (ρ): 15 Ω*cm.
- Resistance: $R = \frac{\rho A_C}{L}$.
- A_C = cross-sectional area of material.
- L = length of material.

Design Principles

- Conductive trace width: 2mm.
- Minimum conductive trace spacing: 2mm.
- Conductive trace height: 2mm.
- Minimum bending angle: 45 degrees.
- Insulating layer thickness: 2mm.
- Insulating layer and Conductive Trace layer are generated separately, not as a single OpenSCAD project.
- The insulating layer should provide 2mm of clearance on each trace boundary
    - i.e., the trace footprint outline is 25.4 mm by 25.4 mm; the insulating layer footprint should be 29.4 mm by 29.4 mm, matching the general shape of the trace footprint outline.
- 4 mm by 4 mm contact pads should be placed at the end of each circuit, with full contact between the end of the trace and the contact pad.
- Contact pads should be given 2mm of spacing on each side from other conductive trace elements.
- Only refer to allowed geometry styles when constructing components

Allowed Geometry Styles

Resistors

- Serpentine

Capacitors

- Interdigitated

Inductors

- Planar

Optimization Goals

N/A

Style Preferences

- The insulating layer serves as a flat platform for conductive traces to be printed on.
- Only create walls in the insulating layer if necessary to isolate distinct trace planes.
- All serpentine bending should be constructed using 45-degree chamfered segments